

Concept Note: Down-to-Earth Data Analysis Using Inverse Theory Hands-On

1. Introduction

Geophysical data analysis lies at the core of modern Earth sciences, enabling researchers to infer subsurface properties from indirect measurements. Inverse theory provides a rigorous mathematical framework for transforming observed geophysical data into meaningful interpretations. This concept note outlines a hands-on training module that integrates linear algebra, SVD, and factor analysis with examples from environmental science, paleontology and petrology and applications to global warming and current climate change.

2. Objectives

- Develop a conceptual and computational understanding of inverse problems in geophysics using Python and MatLab/R.
- Apply linear algebra and probability theory to model data relationships and solve over/under determined systems.
- Implement Singular Value Decomposition (SVD) for inverse problems and factor analysis to identify dominant patterns in multivariate geophysical datasets.
- Use the iLOSCAR code to invert paleogeographical data and compare with current global warming and climate change.

3. Topics Covered

3.1 Introduction to inverse theory: using observations to infer hidden properties.

Forward problem: Given a model (e.g., Earth layers), predict data.

Inverse problem: Given data, find the best model.

We write data as $\mathbf{d} = \mathbf{Gm}$, where $\mathbf{d}$ = observed data, $\mathbf{G}$ = sensitivity matrix (how model affects

data), **m** = model parameters (e.g., layer thickness, velocity)

3.2 Linear Algebra and Matrix Analysis foundation: Matrix operations, rank, null space, eigenvalues and eigenvectors. **Singular Value Decomposition (SVD)** breaks down the matrix **G** into simpler parts. Introduction to over- and under determined problems.

3.3. Probability & Bayes Theory: Posteriors, Likelihoods and Priors, Hypothesis testing, Correlation and Regression, Analysis of Variance applied to ecological data

3.4 Factor Analysis: SVD applied to rock chemistry data taken from a petrology database to deduce the number of the source rocks (called factors) and their composition from observations of the composition of the sediments (called samples)

3.5 Earth's Layers & Reflectivity: Solving the seismological problem of estimating the properties of an interface using measurements of reflection coefficients

3.6 Seismic Travel time Tomography: a widely used method of imaging the Earth's interior by using the travel times of waves propagating through the medium

3.7 Climate Change: analysis of sedimentary data records from the Permian epoch, real-world predictions from the IPCC using the iLOSCAR code, inferences for greenhouse effect

4. Tools & Software

Python: NumPy, SciPy, scikit-learn, **R:** Basic hypothesis testing, regression, ANOVA, **iLOSCAR:** For carbon cycle modeling (available via GitHub)

5. Expected Outcomes

- Proficiency in formulating and solving inverse problems in geophysics.
- Practical experience in applying SVD and factor analysis to real geoscience data.
- Improved interdisciplinary collaboration between mathematics, statistics, hydrology, petrology, environmental science, and geophysics.

6. List of references

- *Geophysical Data Analysis and Inverse Theory with MatLab or Python*, Fifth Edition (2024), William Menke, <https://shop.elsevier.com/books/geophysical-data-analysis-and-inverse-theory-with-matlab-and-python/menke/978-0-443-13794-5> and associated online material.
- *Applied Linear Algebra*, Olver and Shakiban (2018), <https://link.springer.com/book/10.1007/978-3-319-91041-3>
- *Mathematical Methods for Physics and Engineering*, Third Edition (2006), Riley, Hobson and Bence, <https://www.amazon.in/Mathematical-Methods-Physics-Engineering-Comprehensive-ebook/dp/B00AKE1QJU>
- *Geophysical Inverse Theory by John Scales*, Martin Smith and Sven Treitel, Colorado School of Mines, <https://samizdat.mines.edu/geophysical-inverse-theory-by-john-scales-martin-smith-and-sven-treitel/>
- *Singular Value Decomposition* lecture by Royi Avital, <https://speakerdeck.com/royiavital/singular-value-decomposition-why-and-how>
- *A Guided Tour of Mathematical Methods for the Physical Sciences*, Roel Snieder and Kasper van Wijk (2015), <https://www.cambridge.org/core/books/guided-tour-of-mathematical-methods-for-the-physical-sciences/3C5AAC644C2B09F3F17837E703516920>